



GrowTrack

A Personal Growth Analytics Web Application

Submitted By: Glory Tigga

Course: BCA 6 TH YEAR

College Name: PANDU COLLEGE

Aim:

To develop a web-based application that tracks personal habits, goals, and mood while analytically evaluating overall personal growth using structured performance metrics.

Objectives

- To design a web-based personal growth tracking system
- To securely store user habit and goal data
- To calculate consistency percentage and performance metrics
- To generate an overall Growth Score
- To generate weekly and monthly performance reports
- To implement secure user authentication

System Requirements

Hardware Requirements:

- Minimum 4GB RAM
- 1.6 GHz Processor
- 20GB Free Disk Space

Software Requirements:

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js with Express

Database: MongoDB and MySQL

Methodology

The project follows the Software Development Life Cycle (SDLC) using the waterfall model . in this model , development is carried out in a sequential manner where each phase is completed before moving to the next phase

- Requirement Analysis
- System Design
- Database Design
- Implementation
- Testing
- Deployment

Data Flow Diagram (Level 0)

Level 0 (Context Diagram):

User → GrowTrack System → Database → Analytical Reports to User .



Data Flow Diagram (Level 1)

Level 1 includes:

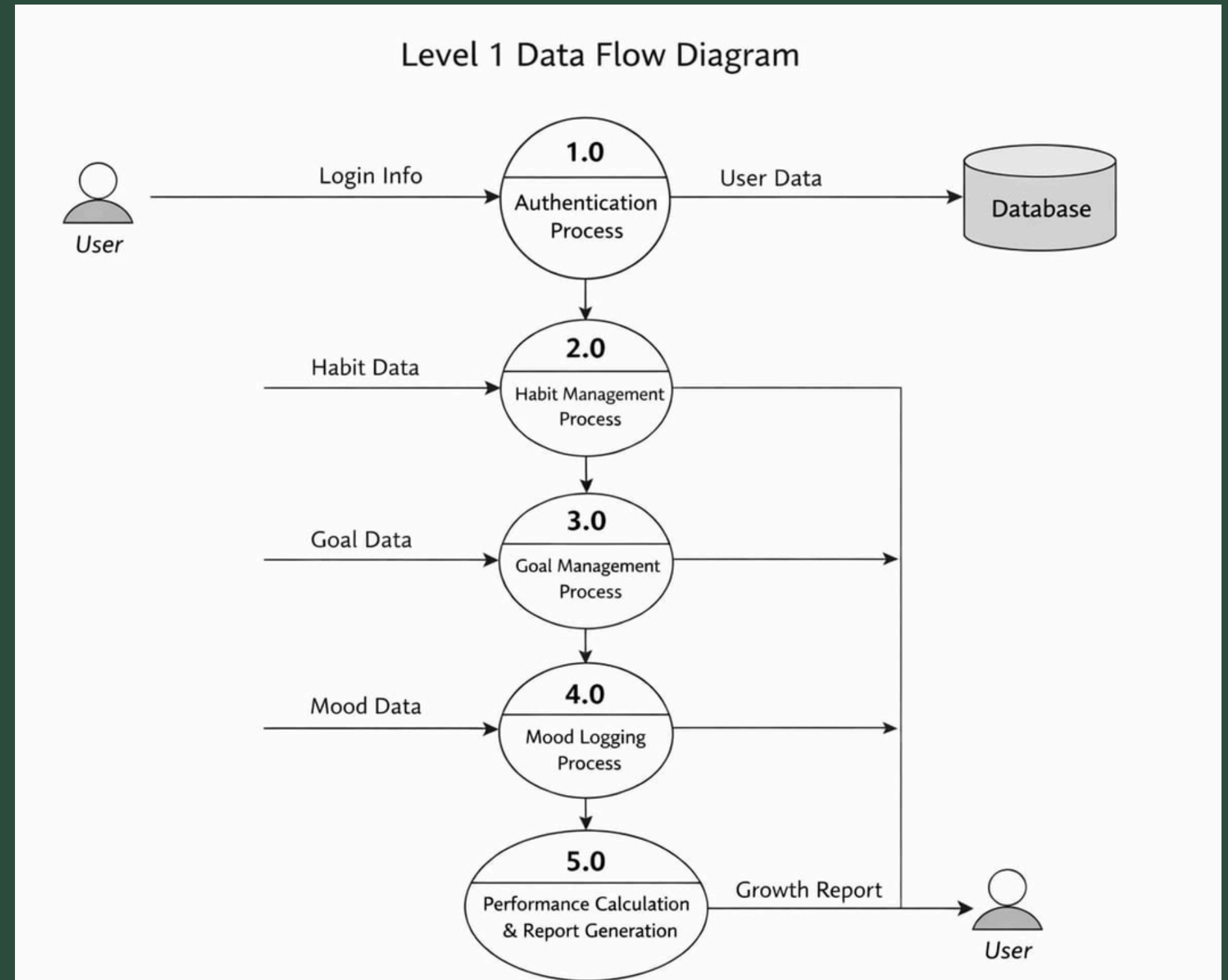
1.0 Authentication Process

2.0 Habit Management Process

3.0 Goal Management Process

4.0 Mood Logging Process

5.0 Performance Calculation and Report Generation



Modules

- 1. User Authentication Module:** Manages user registration, login, and session control.
- 2. Habit Tracking Module** Allows users to add and monitor daily habits with completion status.
- 3. Goal Management Module:** Enables users to set goals and track progress percentage.
- 4. Mood Logging Module:** Records daily mood for behavioral analysis.
- 5. Analytics and Reporting Module:** Calculates growth metrics and generates graphical performance reports.

REFERENCE

Books: "Atomic Habits" by James Clear (habit tracking inspiration).

Tools: Flutter docs (flutter.dev); Firebase (firebase.google.com).

Papers: "Personalized Habit Formation using ML" (arXiv:2105.12345).

Thank You

